

E063

# Introduction to Software Engineering

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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# How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

A food bank asks a volunteer team to build an app for scheduling pickups. The team has two strong Python developers and a laptop donated by a local business. The food bank's director describes the problem in a short phone call. What should the team do first?

- A. Write the test plan.
- B. Start coding in Python since the team already knows it.
- C. Pick a database and draw the system architecture.
- D. Gather and document requirements with the food bank staff.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2.

A clinic's patient portal is being rebuilt by a team that recently moved to a new office. The requirement reads: the search page must return results within 2 seconds for 500 users at once. The old portal was written in 2011. How should this requirement be classified?

- A. Non-functional (performance).
- B. A design constraint on the database.
- C. Functional, because it mentions the search page.
- D. A legacy requirement, because the old portal is from 2011.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3. [TRANSFER — reversed stem / different variable]

Which of these is a functional requirement?

- A. The system is available 99.9% of the time.
- B. Pages load in under 1 second.
- C. A user can reset a forgotten password by email.
- D. All data is encrypted at rest.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4.

A startup is building a new social app for college students. The founders expect to change features every few weeks based on how students react. They have raised \$2 million and rent space downtown. Which approach fits best?

- A. Waterfall, because a \$2 million budget needs a fixed plan.
- B. No process at all, since Agile means no planning.
- C. An Agile approach with short iterations and frequent feedback.
- D. Waterfall, so every feature is fully specified before coding.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

5.

A payroll tool passes every test in its written specification. The team used a well-known testing framework and worked overtime for two weeks. When accountants start using it, they find it cannot handle part-time workers, which they need every day. What failed?

- A. Unit testing, because a unit test must have been skipped.
- B. Validation: the team built the product right, but not the right product.
- C. Verification: the code did not match the specification.
- D. The testing framework, because it missed the defect.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

6. [TRANSFER — reversed stem / different variable]

A reviewer checks that each function in a new module matches its design document line by line. What kind of activity is this?

- A. Verification
- B. Deployment
- C. Acceptance testing
- D. Validation

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7.

A developer on a four-person team writes a test that sends an order from the shopping cart module to the payment module and checks that the payment module receives the correct total. She wrote it on a Friday afternoon using a free test runner. What level of test is this?

- A. Unit test.
- B. Integration test.
- C. System test, because any test that passes data is a system test.
- D. Acceptance test.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8. [TRANSFER — reversed stem / different variable]

A customer's staff run through their real daily tasks on the finished system to decide whether to sign off on it. What level of testing is this?

- A. Unit testing
- B. Regression testing only
- C. Acceptance testing
- D. Integration testing

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

9.

Three developers keep overwriting each other's changes by emailing zip files of the project. The project is written in Java and hosted on a school server with 64 GB of RAM. What practice fixes the core problem?

- A. Move the project to a server with more RAM.
- B. Use a version control system with branches and merges.
- C. Name each zip file with the date.
- D. Have only one developer work at a time, always.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

10.

A popular banking app, used by 2 million customers, must be updated because the phone operating system is removing an old location feature it relies on. The app has had no reported bugs this year. What type of maintenance is this?

- A. No maintenance is needed because there are no bugs.
- B. Perfective maintenance.
- C. Corrective maintenance.
- D. Adaptive maintenance.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

11. [TRANSFER — reversed stem / different variable]

A team rewrites a confusing module so it will be easier to change later. Users will see no difference. What type of maintenance is this?

- A. Adaptive maintenance
- B. Corrective maintenance
- C. Perfective maintenance, because every change improves the product
- D. Preventive maintenance

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

12.

At a mid-sized software company with a ping-pong table and flexible hours, one person builds automated pipelines that test and deploy code, and gets paged when production servers go down. Which role best fits this work?

- A. DevOps or site reliability engineer.
- B. Product manager.
- C. QA test engineer.
- D. UX designer.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer D Concept 1

A food bank asks a volunteer team to build an app for scheduling pickups. ~~The team has two strong Python developers and a laptop donated by a local business.~~ The food bank's director describes the problem in a short phone call. What should the team do first?

**Target word:** first SDLC activity **Decides:** The problem is only loosely described; requirements come first.

|   |                    |                                                                  |
|---|--------------------|------------------------------------------------------------------|
| A | Inverted operation | Tests check requirements, so they come after requirements exist. |
| B | Camouflage         | The team's skills do not change the phase order.                 |
| C | Adjacent category  | That is design, the phase after requirements.                    |
| D | <b>Correct</b>     | You cannot design or build what has not been defined.            |

## 2. Answer A Concept 2

A clinic's patient portal is being rebuilt ~~by a team that recently moved to a new office.~~ The requirement reads: the search page must return results within 2 seconds for 500 users at once. ~~The old portal was written in 2011.~~ How should this requirement be classified?

**Target word:** requirement type **Decides:** It states how fast, not what the system does.

|   |                   |                                                                                        |
|---|-------------------|----------------------------------------------------------------------------------------|
| A | <b>Correct</b>    | A response time under load is a quality attribute.                                     |
| B | Adjacent category | It says nothing about how the database is built.                                       |
| C | Confusion set     | Naming a feature does not make the requirement functional; the measured part is speed. |
| D | Camouflage        | The age of the old system does not classify the requirement.                           |

## 3. Answer C Concept 2 · transfer

Which of these is a functional requirement?

**Target word:** pick the functional one **Decides:** Functional = a behavior the system performs.

|   |                    |                                                |
|---|--------------------|------------------------------------------------|
| A | Confusion set      | Availability is non-functional.                |
| B | Inverted operation | Performance is non-functional.                 |
| C | <b>Correct</b>     | Describes something the system does.           |
| D | Adjacent category  | A security quality constraint, non-functional. |

## 4. Answer C Concept 3

A startup is building a new social app for college students. The founders expect to change features every few weeks based on how students react. ~~They have raised \$2 million and rent space downtown.~~ Which approach fits best?

**Target word:** choose process model **Decides:** Requirements are expected to change often.

|   |                    |                                                                     |
|---|--------------------|---------------------------------------------------------------------|
| A | Camouflage         | Funding does not decide the model; requirement stability does.      |
| B | Absolute language  | Agile still plans, just in shorter cycles.                          |
| C | <b>Correct</b>     | Agile is built to absorb changing requirements.                     |
| D | Inverted operation | Locking requirements up front clashes with planned frequent change. |

## 5. Answer B Concept 4

A payroll tool passes every test in its written specification. ~~The team used a well known testing framework and worked overtime for two weeks.~~ When accountants start using it, they find it cannot handle part-time workers, which they need every day. What failed?

**Target word:** which check failed **Decides:** The spec was met, but the real user need was not.

|          |                    |                                                                |
|----------|--------------------|----------------------------------------------------------------|
| A        | Adjacent category  | Unit tests check code against the spec, which it already met.  |
| <b>B</b> | <b>Correct</b>     | Meeting the spec but missing the need is a validation failure. |
| C        | Inverted operation | The stem says it passed every spec test.                       |
| D        | Camouflage         | The tool is not the problem; the spec left out a need.         |

## 6. Answer A Concept 4 · transfer

A reviewer checks that each function in a new module matches its design document line by line. What kind of activity is this?

**Target word:** name the activity **Decides:** Comparing work to a spec is verification.

|          |                    |                                                                |
|----------|--------------------|----------------------------------------------------------------|
| <b>A</b> | <b>Correct</b>     | It asks whether the product was built right against its spec.  |
| B        | Adjacent category  | Deployment releases software; it does not check it.            |
| C        | Confusion set      | Acceptance is judged by the customer against their criteria.   |
| D        | Inverted operation | Validation checks the real user need, not the design document. |

## 7. Answer B Concept 5

A developer ~~on a four person team~~ writes a test that sends an order from the shopping cart module to the payment module and checks that the payment module receives the correct total. ~~She wrote it on a Friday afternoon using a free test runner.~~ What level of test is this?

**Target word:** testing level **Decides:** The test checks two modules working together.

|          |                   |                                                                                     |
|----------|-------------------|-------------------------------------------------------------------------------------|
| A        | Confusion set     | A unit test checks one piece in isolation.                                          |
| <b>B</b> | <b>Correct</b>    | It checks the interface between two modules.                                        |
| C        | Absolute language | System tests cover the whole application, not one interface.                        |
| D        | Adjacent category | Acceptance tests are judged against customer criteria, usually on the whole system. |

## 8. Answer C Concept 5 · transfer

A customer's staff run through their real daily tasks on the finished system to decide whether to sign off on it. What level of testing is this?

**Target word:** testing level from who judges **Decides:** The customer decides sign-off.

|          |                    |                                                                               |
|----------|--------------------|-------------------------------------------------------------------------------|
| A        | Inverted operation | The smallest level, run by developers, is the opposite end.                   |
| B        | Adjacent category  | Regression re-checks old features after changes; it is not the sign-off step. |
| <b>C</b> | <b>Correct</b>     | The customer checks the system against their own criteria.                    |
| D        | Confusion set      | Checks module interfaces, not customer sign-off.                              |

## 9. Answer B Concept 6

Three developers keep overwriting each other's changes by emailing zip files of the project. ~~The project is written in Java and hosted on a school server with 64 GB of RAM.~~ What practice fixes the core problem?

**Target word:** fix for conflicting changes **Decides:** Parallel changes need tracked history and merging.

|          |                   |                                                          |
|----------|-------------------|----------------------------------------------------------|
| A        | Camouflage        | Hardware does not fix overwritten work.                  |
| <b>B</b> | <b>Correct</b>    | Tracks every change and merges parallel work safely.     |
| C        | Dropped element   | Still no merging or change tracking.                     |
| D        | Absolute language | Avoids conflicts by removing teamwork, not solving them. |

## 10. Answer D Concept 7

A popular banking app, ~~used by 2 million customers,~~ must be updated because the phone operating system is removing an old location feature it relies on. The app has had no reported bugs this year. What type of maintenance is this?

**Target word:** maintenance type **Decides:** The change responds to a changed environment.

|          |                    |                                                                 |
|----------|--------------------|-----------------------------------------------------------------|
| A        | Inverted operation | Maintenance covers more than bug fixes.                         |
| B        | Adjacent category  | Perfective adds or improves features users want.                |
| C        | Confusion set      | Corrective fixes defects; the stem says there are none.         |
| <b>D</b> | <b>Correct</b>     | Adjusting to a change in the operating environment is adaptive. |

## 11. Answer D Concept 7 · transfer

A team rewrites a confusing module so it will be easier to change later. Users will see no difference. What type of maintenance is this?

**Target word:** maintenance type from purpose **Decides:** Purpose is to prevent future problems.

|          |                   |                                                                         |
|----------|-------------------|-------------------------------------------------------------------------|
| A        | Adjacent category | No outside environment changed.                                         |
| B        | Confusion set     | No defect is being fixed.                                               |
| C        | Absolute language | Perfective targets features users see; this has no user-visible change. |
| <b>D</b> | <b>Correct</b>    | Improves maintainability to head off future issues.                     |

## 12. Answer A Concept 8

At a mid-sized software company ~~with a ping pong table and flexible hours,~~ one person builds automated pipelines that test and deploy code, and gets paged when production servers go down. Which role best fits this work?

**Target word:** role from duties **Decides:** Automating delivery and keeping production running.

|          |                   |                                                                  |
|----------|-------------------|------------------------------------------------------------------|
| <b>A</b> | <b>Correct</b>    | Pipelines and production uptime are the core of DevOps/SRE work. |
| B        | Confusion set     | Decides what to build and why, not pipelines.                    |
| C        | Adjacent category | Focuses on verifying quality, not deployment and on-call.        |
| D        | Dropped element   | Ignores every technical duty in the stem.                        |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 2             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 0             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 10            |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 7             |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 4             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 8             |             |
| Camouflage         | Answers a stem fact that was never the question.             | 5             |             |

# Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                           | Rule                                                                                                                                                                        | Trap                                                                                         | Tell                                               |
|---|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------|
| 1 | <b>SDLC phase order</b>                           | Requirements come before design, design before implementation, then testing, deployment, and maintenance.                                                                   | Starting to code before the problem is defined.                                              | “What should the team do first/next?”              |
| 2 | <b>Functional vs. non-functional requirements</b> | Functional requirements say what the system does; non-functional requirements say how well it does it (speed, security, usability, reliability).                            | Calling any requirement with a feature name functional, even when it states a response time. | A number about speed, uptime, or capacity.         |
| 3 | <b>Waterfall vs. Agile fit</b>                    | Plan-driven (Waterfall) fits stable, well-understood, fixed requirements; Agile fits requirements that are uncertain or likely to change.                                   | Treating Agile as always better regardless of context.                                       | How settled the requirements are.                  |
| 4 | <b>Verification vs. validation</b>                | Verification asks “did we build it right?” (meets the spec); validation asks “did we build the right thing?” (meets the user’s real need).                                  | Swapping the two.                                                                            | Spec vs. user need.                                |
| 5 | <b>Levels of testing</b>                          | Unit tests check one piece alone; integration tests check pieces working together; system tests check the whole; acceptance tests check it against the customer’s criteria. | Calling a test that crosses two modules a unit test.                                         | How many parts the test touches and who judges it. |
| 6 | <b>Version control and branching</b>              | Commit small, related changes; use branches to isolate work and merge through review so the main line stays working.                                                        | Everyone committing straight to main, or emailing zipped copies.                             | Several people changing the same code base.        |
| 7 | <b>Types of maintenance</b>                       | Corrective fixes defects; adaptive adjusts to a changed environment; perfective improves or adds features; preventive reduces future problems.                              | Calling every post-release change a “bug fix.”                                               | Why the change is being made.                      |
| 8 | <b>Software engineering roles</b>                 | Developers build; QA/test engineers verify quality; DevOps/SRE engineers automate delivery and keep systems running; product managers decide what to build and why.         | Assuming the product manager writes code or QA sets priorities.                              | The task described, not the job title.             |

**Course scope (public description):** This course guides learners toward understanding what Software Engineering means, how it applies in the real world, and the landscape of careers in the field. It also includes a little review of programming while being appropriate for learners with little to no background in software engineering.